Determination of SG and Absorption for Coarse Aggregate (ASTM C127)

- Sample of a coarse aggregate is soaked for 24 hours and rolled in a large absorbent cloth to remove all visible surface moisture to achieve the SSD condition
- Weighed suspended in water, W_w
- The sample is then dried to SSD condition again and weighed, W_{SSD}
- Finally, the sample is bone dried and weighed, W_{OD}

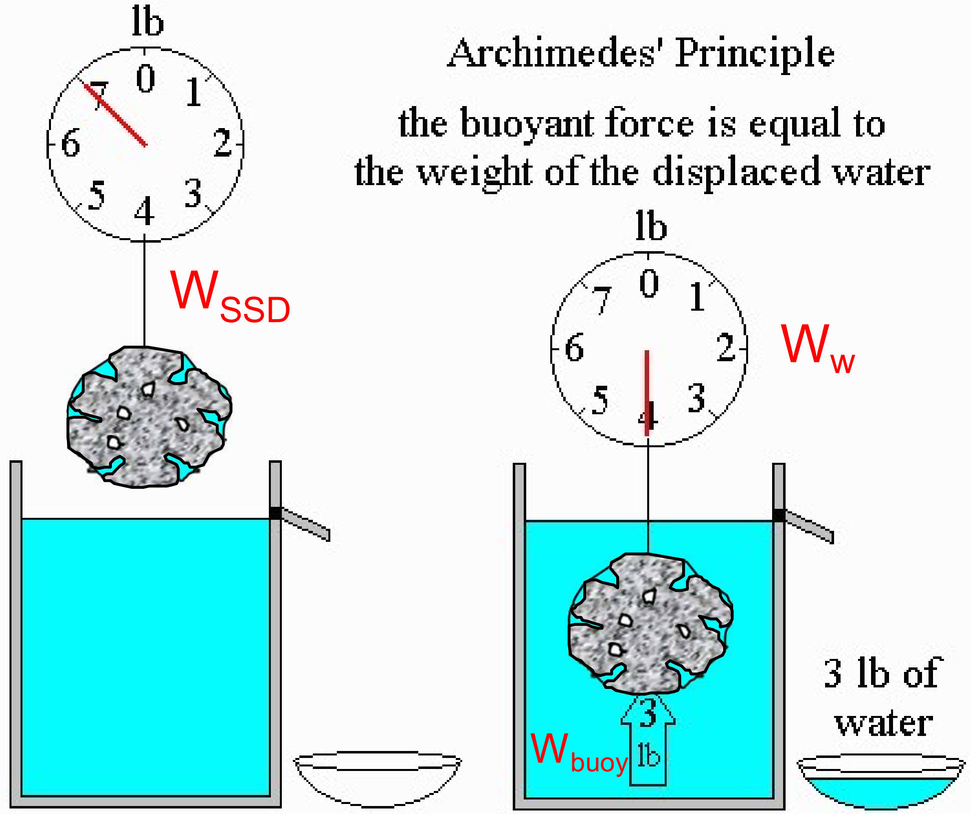
http://youtu.be/NlN3OgiMcms

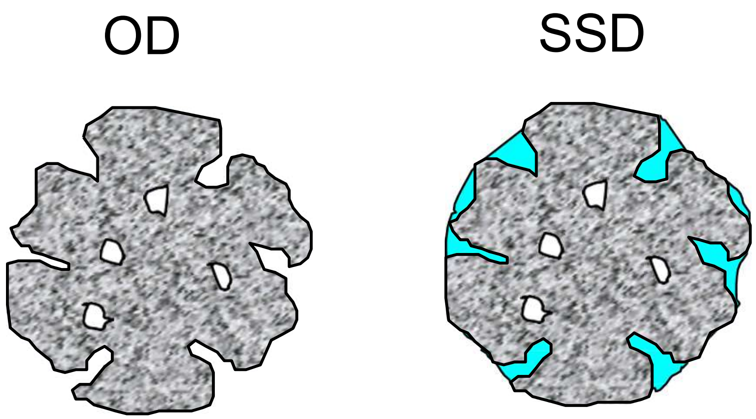
$$W_{buoy} = V_{SSD}\rho_w$$

$$W_w = W_{SSD} - W_{buoy} = W_{SSD} - V_{SSD}\rho_w$$

$$V_{SSD}\rho_w = W_{SSD} - W_w$$

$$\therefore BSG_{SSD} = \frac{W_{SSD}}{V_{SSD}\rho_w} = \frac{W_{SSD}}{W_{SSD} - W_w}$$





$$BSG_{SSD} = \frac{W_{SSD}}{V_{SSD}\rho_w} = \frac{W_{SSD}}{W_{SSD} - W_w}$$

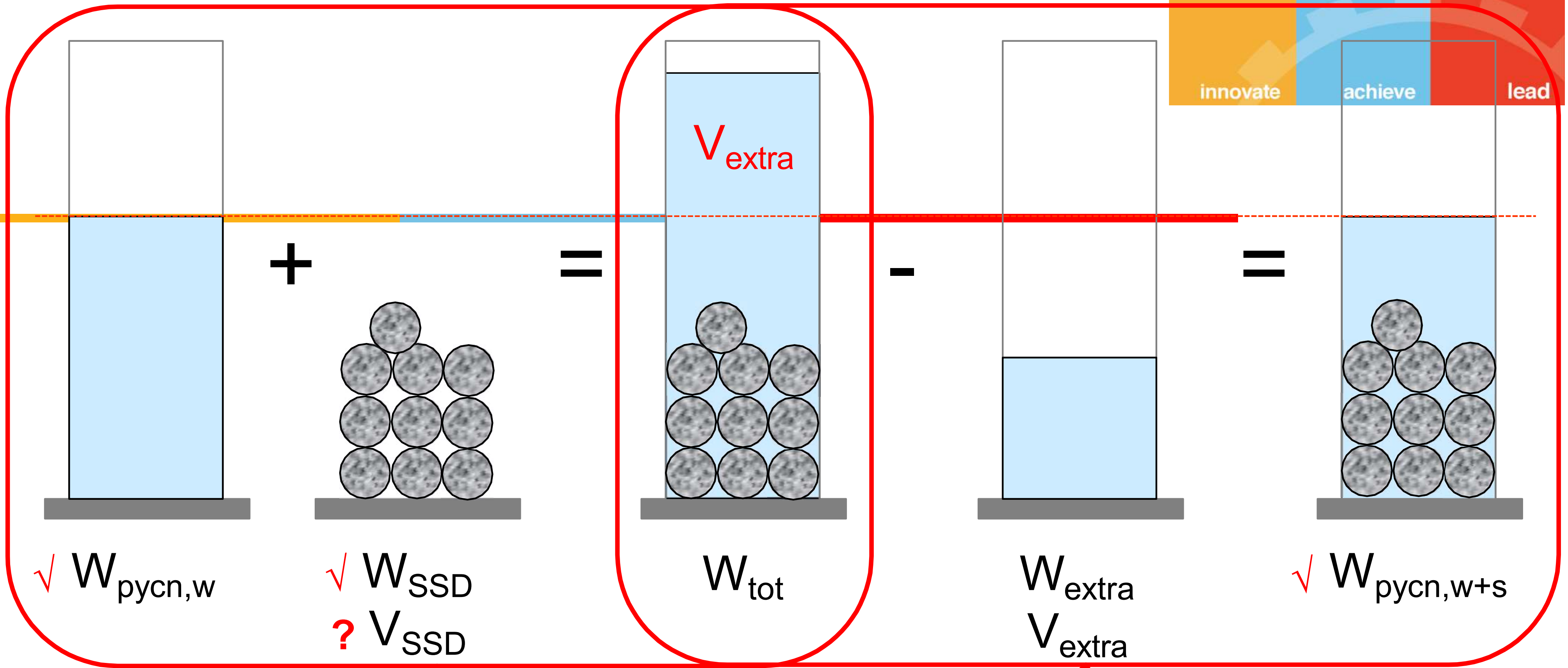
$$BSG_{OD} = \frac{W_{OD}}{V_{SSD}\rho_w} = \frac{W_{OD}}{W_{SSD} - W_w}$$

$$ASG = \frac{W_{OD}}{(V_{SSD} - V_p)\rho_w} = \frac{W_{OD}}{(W_{SSD} - W_w) - W_p} = \frac{W_{OD}}{W_{OD} - W_w}$$

$$Absorption\ (\%) = \frac{W_{SSD} - W_{OD}}{W_{OD}} \times 100$$

Determination of SG and Absorption for Fine Aggregate (ASTM C128)

- Sample of a fine aggregate is soaked for 24 hours to achieve SSD condition
- A 500-g SSD sample (W_{SSD}) is placed in **pycnometer**, (a constant volume flask)
- Water is added to the constant volume mark on the pycnometer and weighed, $W_{pyc,w+s}$
- Remove the sample and water from the pycnometer and re-fill the pycnometer with water only to the same volume mark and weighed, $W_{pyc,w}$
- The sample is then bone dried and weighed, W_{OD}



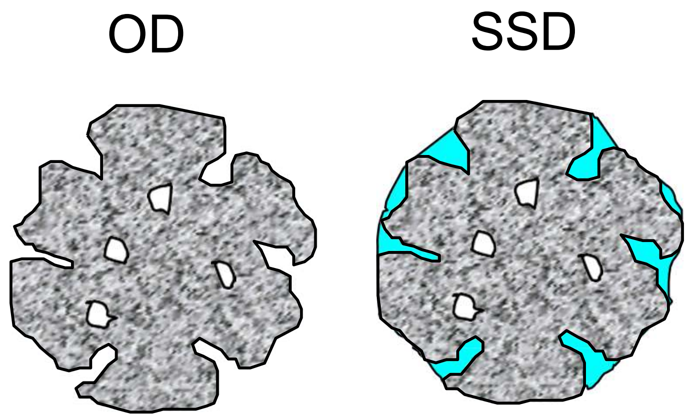
$$V_{SSD} = V_{extra}$$

$$V_{SSD}\rho_w = V_{extra}\rho_w = W_{extra} = W_{tot} - W_{pycn,w+s} = W_{pycn,w} + W_{SSDs} - W_{pycn,w+s}$$

$$\therefore BSG_{SSD} = \frac{W_{SSD}}{V_{SSD}\rho_w} = \frac{W_{SSD}}{W_{pycn,w} + W_{SSD} - W_{pycn,w+s}}$$

$$BSG_{SSD} = \frac{W_{SSD}}{V_{SSD}\rho_w} = \frac{W_{SSD}}{W_{pycn,w} + W_{SSD} - W_{pycn,w+s}}$$

$$BSG_{OD} = \frac{W_{OD}}{V_{SSD}\rho_w} = \frac{W_{OD}}{W_{pycn,w} + W_{SSD} - W_{pycn,w+s}}$$



$$ASG = \frac{W_{OD}}{(V_{SSD} - V_p)\rho_w} = \frac{W_{OD}}{(W_{pycn,w} + W_{SSD} - W_{pycn,w+s}) - W_p}$$

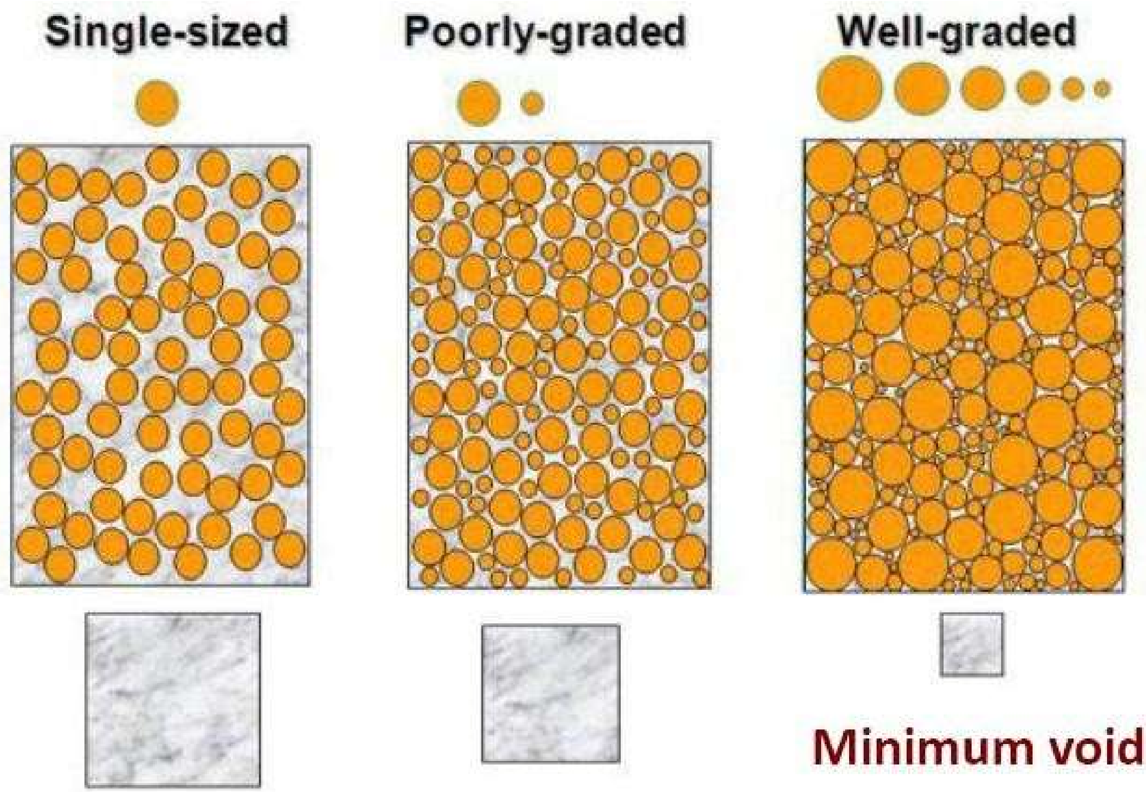
$$= \frac{W_{OD}}{W_{pycn,w} + W_{OD} - W_{pycn,w+s}}$$

$$Absorption\ (\%) = \frac{W_{SSD} - W_{OD}}{W_{OD}} \times 100$$

[Pycnometer used for FA Specific Gravity]

Gradation

- Gradation is an important attribute to produce economical concrete
  - Max. density (i.e. min. voids)
  - Min. cement content



Void content of aggregate:

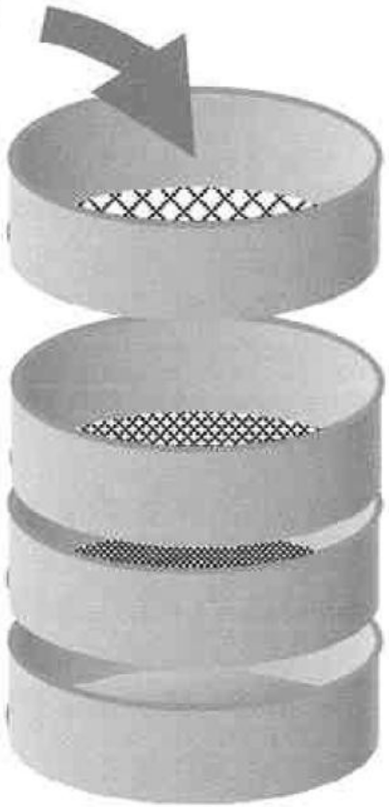
: Particle size distribution

: Maximum aggregate size

Sieve Analysis for Gradation

- **Gradation**: Particle size distribution of aggregate
- **Sieve Analysis**: Process of dividing aggregate into fractions of same particle size in order to determine gradation of aggregate
  - Standard coarse sieves: 37.5mm; 19mm; 9.5mm; 4.75mm
  - Standard fine sieves: 4.75mm; 2.36mm; 1.18mm; 0.60mm; 0.30mm; 0.15mm
- **Grading Curve**: Usually described by the cumulative percentage of aggregates that either pass through or retained by a specific sieve size

Aggregates are poured into the test sieves, which are placed in order of decreasing aperture size from top to bottom.



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Sieve Analysis for Fine Aggregate Sample & Fineness Modulus

Sieve size (mm)	No.	Weight retained (g)	% retained	Cumulative % retained	Cumulative % passing
4.75	4	0	0	0	100
2.36	8	241.9	11.9	11.9	88.1
1.18	16	388.9	19.1	31.0	69.0
0.60	30	505.5	24.9	55.9	44.1
0.30	50	543.4	26.7	82.6	17.4
0.15	100	340.8	16.8	99.4	0.6
Pan		11.3	0.6	100	0
Total		2031.8	100		

Fineness Modulus

- A measure of gradation fineness
- FM = ∑ (Cumulative % retained on standard sieves up to 0.15 mm) / 100

$$FM = \sum R_i \Big/ 100$$

- FM cannot be representative of a distribution, i.e. two different grading curves can have same FMs
- Higher FM, coarser aggregate
- Lower FM is not economical

Sieve Analysis for Fine Aggregate Sample & Fineness Modulus

Sieve size (mm)	No.	Weight retained (g)	% retained	Cumulative % retained	Cumulative % passing
4.75	4	0	0	0	100
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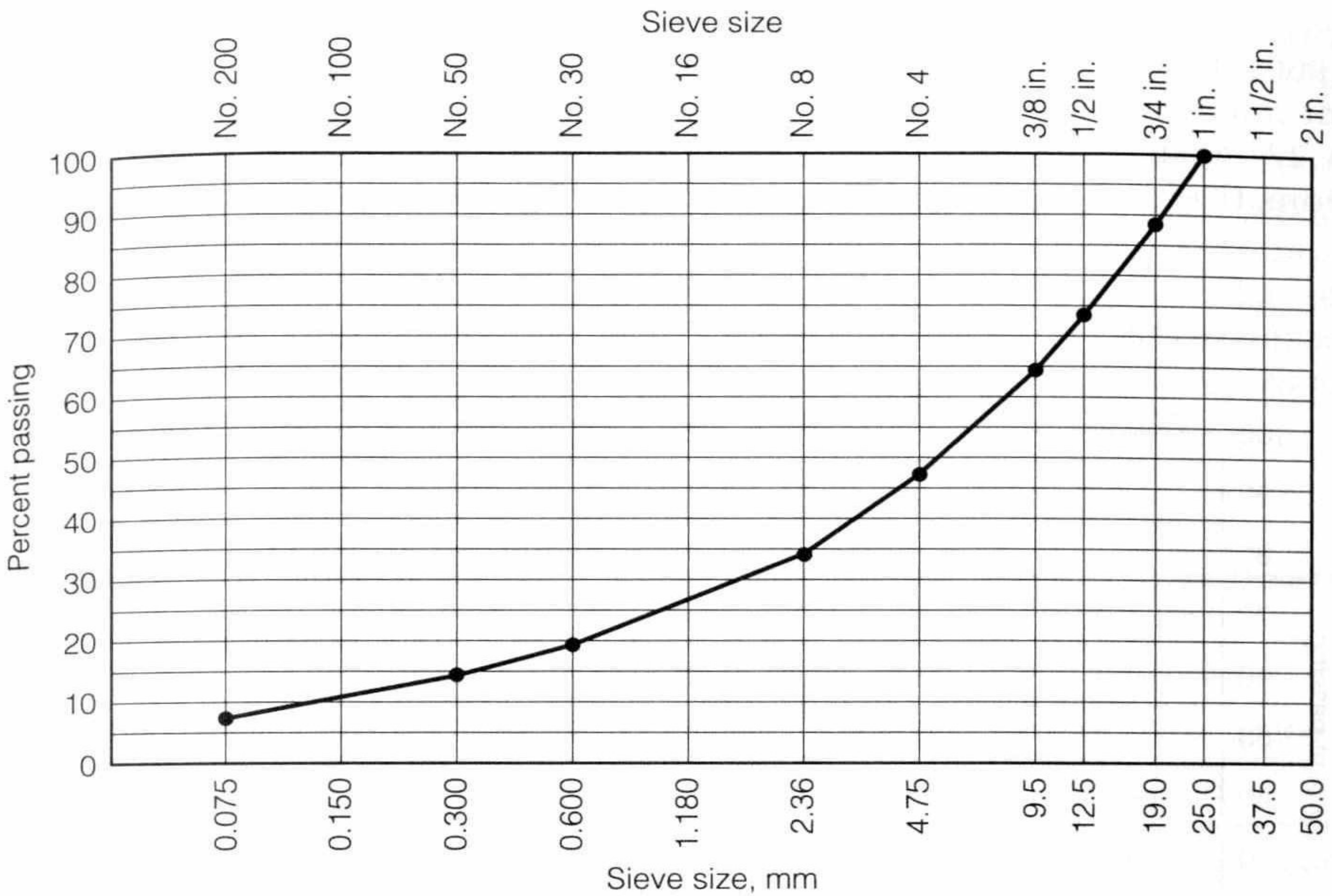
0.60	30	505.5	24.9	55.9	44.1
0.30	50	543.4	26.7	82.6	17.4
0.15	100	340.8	16.8	99.4	0.6
Pan		11.3	0.6	100	0
Total		2031.8	100		

$$FM = \frac{11.9 + 31.0 + 55.9 + 82.6 + 99.4}{100} = \frac{280.8}{100} = 2.81$$

Semi Log Grading Graph



Example

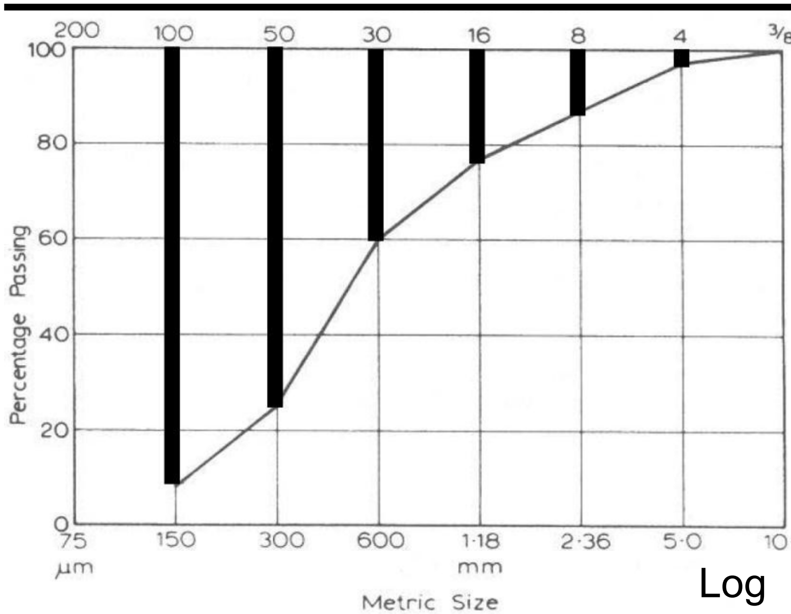
- The following results were obtained from the sieve analysis of fine aggregate sample

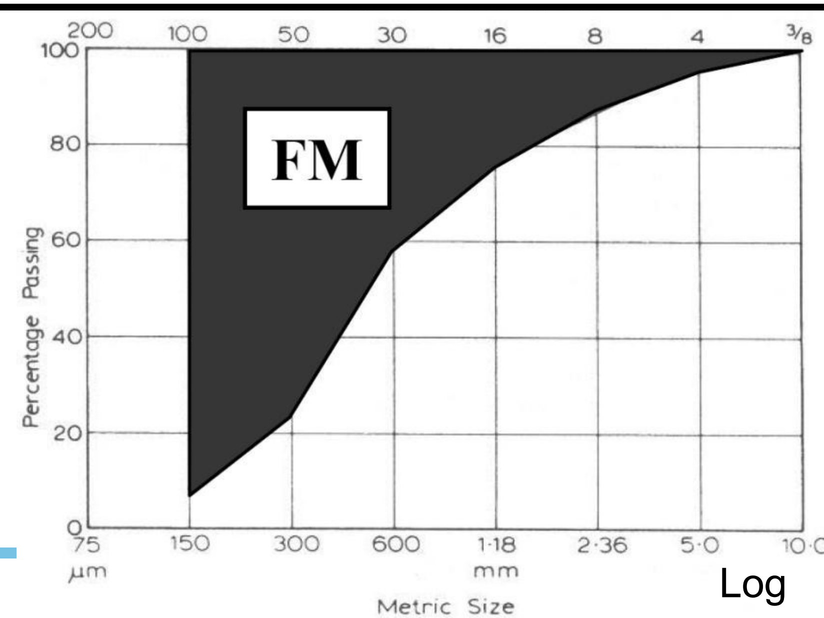
Sieve size (mm)	9.5	4.75	2.36	1.18	0.60	0.30	0.15	Pan
Retained (g)	0	6	31	30	59	107	53	21

- Draw the grading curve for the fine aggregate
- Determine the fineness modulus of the fine aggregate

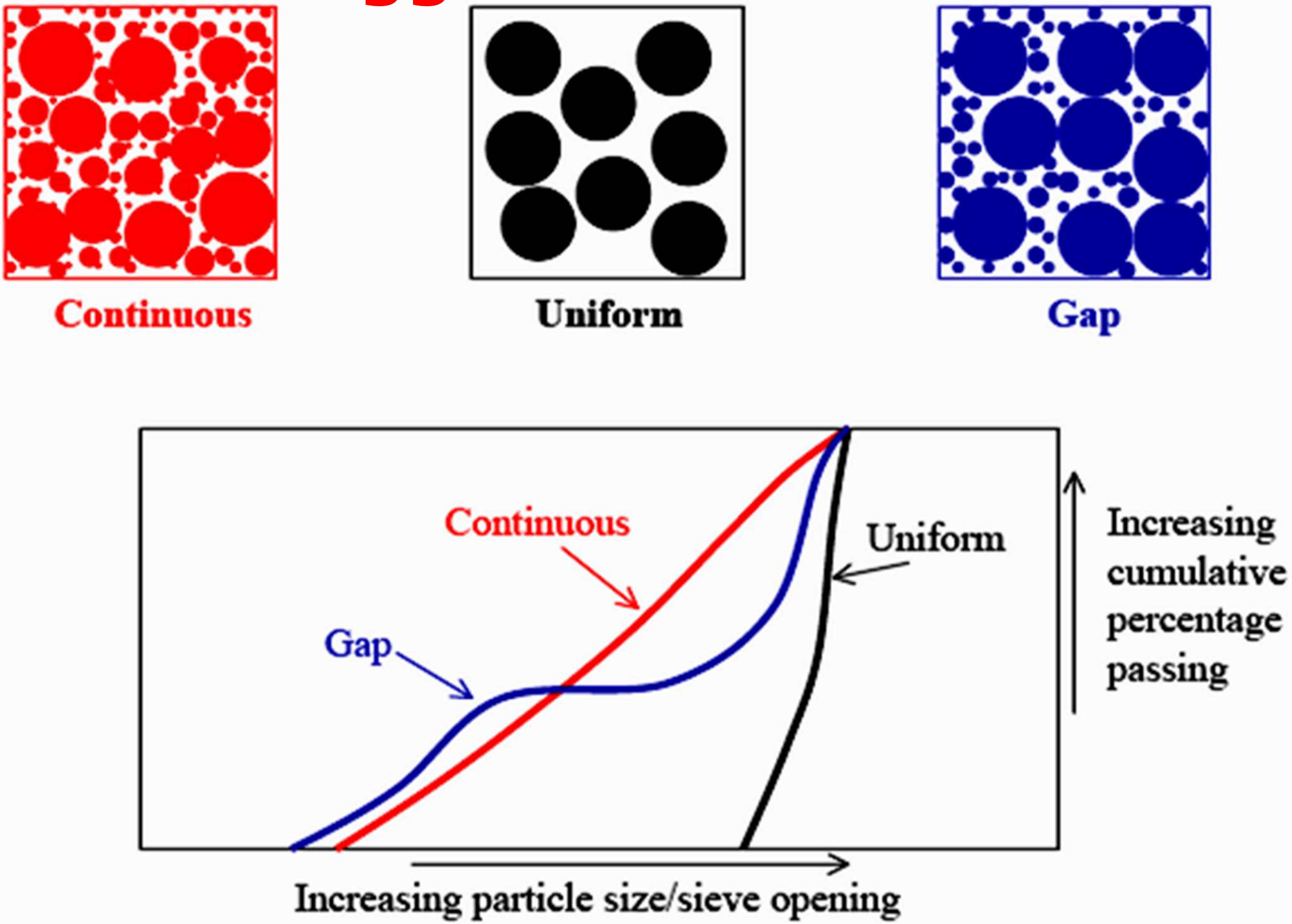
Solution

Sieve size BS (ASTM)	Mass retained, g	% retained	Cumulative % retained	Cumulative % passing
9.50 mm (3/8 in.)	0	0.0	0	100
4.75 mm (No. 4)	6	2.0	2	98
2.36 mm (No. 8)	31	10.1	12	88
1.18 mm (No. 16)	30	9.8	22	78
600 μm (No. 30)	59	19.2	41	59
300 μm (No. 50)	107	34.9	76	24
150 μm (No. 100)	53	17.3	93	7
< 150 μm	21	6.8	---	---
Total	307		246	
FM			2.46	





Schematic Representations of Different Agg. Gradation



Types of Gradation

Continuous (Well graded, dense)

- Has a good mix of all particle sizes which means the aggregates use most of the volume and less cement is needed

One-size gradation (Uniform)

- All same size = nearly vertical curve

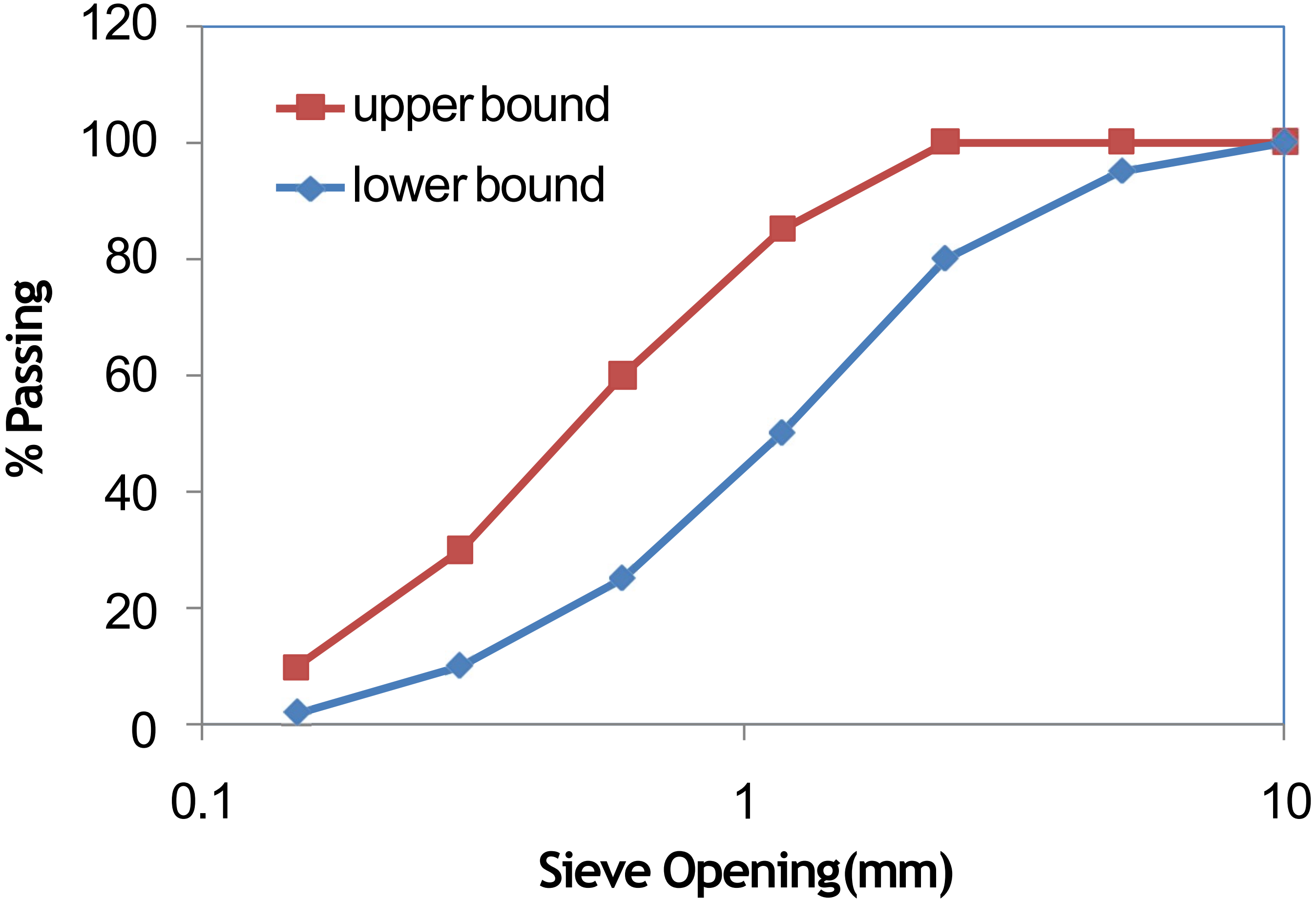
Gap-graded

- Missing some sizes = nearly horizontal section of curve

Open-Graded

- Missing small aggregates which fill in holes between larger ones
- Lower part of curve is skewed toward large sizes

General Grading Requirement for Fine Aggregate (ASTM C33)



Sieve	Percent Passing
9.5 mm (3/8)	100
4.75 mm (No. 4)	95–100
2.36 mm (No. 8)	80–100
1.18 mm (No. 16)	50–85
0.60 mm (No. 30)	25–60
0.30 mm (No. 50)	10–30
0.15 mm (No. 100)	2–10

General Grading Requirement for Coarse Aggregate (ASTM C33)

Sieve size	Percentage by mass passing sieves		
	Nominal size of graded aggregate		
mm	37.5 to 4.75 mm	19.0 to 4.75 mm	12.5 to 4.75 mm
75.0	---	---	---
63.0	---	---	---
50.0	100	---	---
37.5	95-100	---	---
25.0	---	100	---
19.0	35-70	90-100	100
12.5	---	---	90-100
9.5	10-30	20-55	40-70
4.75	0-5	0-10	0-15
2.36	---	0-5	0-5